

AN ESTIMATION PROCEDURE IN CATEGORICAL DATA ANALYSIS

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Summary: A general approach to the estimation of parameters in categorical data analysis, which makes use of properties of singular normal variates, is considered. Various examples, including log-linear models, are given to illustrate the procedure.

1. INTRODUCTION

Since the 1960's various approaches have been followed and statistical models developed for the analysis of categorical data. These models imply different estimation procedures and computing algorithms. The present approach, where the frequency vector is assumed to be approximately normal and which makes use of properties of singular normal variates, attempts to link the models and techniques more closely to the standard normal-theory approach. Provision is made for constraints such as symmetry, fixed subtotals or weighted data.

Estimators of the expected cell frequencies are obtained making use of the approximate normal likelihood. These correspond to the minimum modified chi square (MMCS) estimators obtained by Grizzle and Williams (1972, p.150) in the case of a non-singular covariance matrix of the estimated constraints involved. The estimators are also best asymptotically normal (BAN) estimates, as defined by Neyman (1949).

In order to illustrate the general procedure, an example of a symmetry model and of a binomial response model are presented. The corresponding results for the log-linear model are given, and the special case of a 2^3 contingency table considered. A further three-way example for the comparison of estimation procedures is included. Finally it is also shown how the procedure can be applied for the analysis of frequency structures.

The present approach attempts to simplify modelling and estimation in categorical data analysis.

2. A GENERAL ESTIMATION PROCEDURE

Theorem 1. Let $\underline{x} : p \times 1$ be a MLE (maximum likelihood estimate) for $\underline{\mu}$, where $\underline{x}$ has a multivariate normal, $N(\underline{\mu}, \Sigma)$, distribution for Σ known. The MLE of $\underline{\mu}$, under the restriction $G\underline{\mu} = \underline{c}$, for any known matrix G and known vector $\underline{c}$, is the MLE of the mean of the conditional distribution of $\underline{x}$, given that $G\underline{x} = \underline{c}$, i.e.

$$(2.1) \quad \hat{\underline{\mu}}_{\underline{c}} = \underline{x} + (G\Sigma)'(G\Sigma G')^*(\underline{c} - G\underline{x})$$

where $(G\Sigma G')^*$ is any generalised inverse of $G\Sigma G'$. Furthermore, $\hat{\underline{\mu}}_{\underline{c}}$ is distributed as $N(\underline{\mu}_{\underline{c}}, \Sigma_{\underline{c}})$, with

$$(2.2) \quad \begin{aligned} \underline{\mu}_{\underline{c}} &= \underline{\mu} + (G\Sigma)'(G\Sigma G')^*(\underline{c} - G\underline{\mu}) \\ \Sigma_{\underline{c}} &= \Sigma - (G\Sigma)'(G\Sigma G')^*G\Sigma \end{aligned}$$

and $\text{rank}(\Sigma_{\underline{c}}) = \text{rank}(\Sigma) - \text{rank}(G\Sigma)$.

Proof: We first assume that Σ is non-singular. The distribution of the random vector

$$\begin{pmatrix} \underline{x} \\ G\underline{x} \end{pmatrix} = \begin{pmatrix} I \\ G \end{pmatrix} \underline{x}$$

is normal with mean and covariance matrix:

$$\begin{pmatrix} \underline{\mu} \\ G\underline{\mu} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \Sigma & (G\Sigma)' \\ G\Sigma & G\Sigma G' \end{pmatrix}.$$

The conditional distribution of $\underline{x}$, given $G\underline{x} = \underline{c}$, is then normal with mean and covariance matrix given by (2.2). The maximum of the density function

$$f(\underline{x}) = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} (\underline{x} - \underline{\mu})' \Sigma^{-1} (\underline{x} - \underline{\mu}) \right\}$$

of $\underline{x}$, given that $G\underline{x} = \underline{c}$, is obtained at the mean of the conditional distribution of $\underline{x}$, given that $G\underline{x} = \underline{c}$, i.e. at

$$\underline{x} = \underline{\mu} + (G\Sigma)'(G\Sigma G')^*(\underline{c} - G\underline{\mu}).$$

Since $f(\underline{x})$ is a symmetric function in $\underline{x}$ and $\underline{\mu}$, the roles of $\underline{x}$ and $\underline{\mu}$ may be interchanged, leading to (2.4) as the maximum point of the corresponding likelihood function. The expression (2.2) follows directly, noting that $\hat{\underline{\mu}}_{\underline{c}}$ is a linear transformation of the elements of $\underline{x}$. Since different sets of linear restrictions may be imposed sequentially, Theorem 1 is also true for Σ singular.

Corollary 1. If Σ is known, (2.1) gives an explicit expression for the restricted MLE of $\underline{\mu}$. However if Σ is a known function of $\underline{\mu}$, indicated by $\Sigma(\underline{\mu})$, the restricted MLE of $\underline{\mu}$ can be obtained from (2.1) iteratively. If $\hat{\underline{\mu}}_{c,i}$ is the value of the restricted MLE at the i -th iteration, then:

$$(2.3) \quad \hat{\underline{\mu}}_{c,i} = \hat{\underline{\mu}}_{c,i-1} + [G.\Sigma(\hat{\underline{\mu}}_{c,i-1})]'[G.\Sigma(\hat{\underline{\mu}}_{c,i-1}).G']^*(\underline{c} - G.\hat{\underline{\mu}}_{c,i-1})$$

where $\hat{\underline{\mu}}_{c,0} = \underline{x}$.

As can be expected, the distribution of the restricted MLE of $\underline{\mu}$ is the same as the distribution of $\underline{x}$, given that $G\underline{x} = \underline{c}$.

Consider a frequency vector $\underline{f}$ which is assumed to be approximately $N(\underline{F}, \Sigma)$. The elements of $\underline{F}$ can be expected frequencies in any contingency table or expected frequencies according to some other specified sampling scheme. The parameters $\underline{F}$ and Σ are dependent on the cell probabilities and the sampling scheme.

Any constraints on the elements of $\underline{F}$, including those specified by different hypotheses in a log-linear or logit model, can be expressed in the form: $\underline{g}(\underline{F}) = \underline{c}$, where $\underline{g}$ is a vector valued function of $\underline{F}$, satisfying the usual regularity conditions and $\underline{c}$ a known constant vector. $\underline{g}(\underline{f})$ is assumed to have an approximate normal distribution. Let

$$G = \frac{\partial \underline{g}(\underline{F})}{\partial \underline{F}}$$

be the derivative of $\underline{g}(\underline{F})$ with respect to $\underline{F}$. The problem considered here is that of estimating $\underline{F}$ under these constraints, if the estimator $\underline{f}$ of $\underline{F}$ under the full model, i.e. without the constraints, is known. The joint distribution of $\begin{pmatrix} \underline{f} \\ \underline{g}(\underline{f}) \end{pmatrix}$, making use of the delta method (c.f. Bishop, Fienberg and Holland (1975)), is again approximately normal, with mean $\begin{pmatrix} \underline{F} \\ \underline{g}(\underline{F}) \end{pmatrix}$ and covariance matrix

$$\frac{\partial \begin{pmatrix} \underline{F} \\ \underline{g} \end{pmatrix}}{\partial \underline{F}} \Sigma \left(\frac{\partial \begin{pmatrix} \underline{F} \\ \underline{g} \end{pmatrix}}{\partial \underline{F}} \right)' = \begin{pmatrix} \Sigma & (G\Sigma)' \\ G\Sigma & G\Sigma G' \end{pmatrix}.$$

The conditional distribution of $\underline{f}$, given that $\underline{g}(\underline{f}) = \underline{c}$, is therefore approximately normal with expected value

$$(2.4) \quad \underline{F}_c = E(\underline{f} / \underline{g}(\underline{f}) = \underline{c}) = \underline{F} + (G\Sigma)'(G\Sigma G')^*(\underline{c} - \underline{g}(\underline{F}))$$

and covariance matrix

$$(2.5) \quad \Sigma_c = \Sigma - (G\Sigma)'(G\Sigma G')^*G\Sigma.$$

The rank of the conditional covariance matrix is given by

$$(2.6) \quad \text{rank}(\Sigma_c) = \text{rank}(\Sigma) - \text{rank}(G\Sigma).$$

Since Σ is a function of $\underline{F}$, the estimator $\underline{f}_c$ of $\underline{F}_c$ in (2.4) must be determined iteratively, given the estimator $\underline{f}$ of $\underline{F}$, as explained in Theorem 1 and Corollary 1.

It is interesting to note that the multinomial and product multinomial can be seen as special models of the Poisson model, given the total frequency fixed or certain marginal totals fixed. For example, for Poisson sampling the covariance matrix is a diagonal matrix with the expected frequencies on the diagonal, denoted as $\Sigma = \text{diag}(\underline{F}) = D_F$. The total N fixed can be specified as $\underline{g}(\underline{F}) = \underline{1}'\underline{F} = N$, where $\underline{1}$ is a column vector of ones. Therefore the matrix $G = \underline{1}'$, $G\Sigma = \underline{1}'D_F = \underline{F}'$ and $G\Sigma G' = N$, so that the conditional covariance matrix reduces to $\Sigma_c = D_F - \frac{1}{N}\underline{F}\underline{F}'$, which is simply the multinomial covariance matrix.

3. A SYMMETRY MODEL

A good example of a special case of the general model considered in Section 2 is obtained by considering the example in Haberman (1978, p.25), where frequencies appear in a frequency table in an approximately symmetric way. The example is represented in Table 3.1.

TABLE 3.1
SELF-CLASSIFICATION BY SOCIAL CLASS

<i>Lower class</i>	72
<i>Working class</i>	714
<i>Middle class</i>	655
<i>Upper class</i>	41
<i>Total</i>	1 482

The following set of constraints specifies a symmetry model:

$$(3.1) \quad \underline{g}(\underline{F}) = \begin{pmatrix} 1 & 0 & +0 & -1 \\ 0 & 1 & -1 & +0 \end{pmatrix} \underline{F} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

In the case of Poisson sampling it then follows that

$$\begin{aligned}
 (3.2) \quad \Sigma &= \text{diag}(\underline{F}) = D_F \\
 G &= \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix} \\
 G\Sigma &= \begin{pmatrix} F_1 & 0 & 0 & -F_4 \\ 0 & F_2 & -F_3 & 0 \end{pmatrix} \\
 G\Sigma G' &= \begin{pmatrix} F_1 + F_4 & 0 \\ 0 & F_2 + F_3 \end{pmatrix}.
 \end{aligned}$$

From (2.4)

$$(3.3) \quad \underline{F}_c = \begin{pmatrix} \frac{2}{1/F_1 + 1/F_4} \\ \frac{2}{1/F_2 + 1/F_3} \\ \frac{2}{1/F_2 + 1/F_3} \\ \frac{2}{1/F_1 + 1/F_4} \end{pmatrix}.$$

The estimated frequency under symmetry of the first and fourth cells is the harmonic mean of the observed first and fourth cell frequencies and similarly for the second and third cells. In this example symmetry is obtained with the first application of (2.4), and no iteration is needed. The conditional covariance matrix, from (2.5), is given by:

$$(3.4) \quad \Sigma_c = D_F - \begin{pmatrix} \frac{F_1^2}{F_1 + F_4} & 0 & 0 & -\frac{F_1 F_4}{F_1 + F_4} \\ 0 & \frac{F_2^2}{F_2 + F_3} & -\frac{F_2 F_3}{F_2 + F_3} & 0 \\ 0 & -\frac{F_2 F_3}{F_2 + F_3} & \frac{F_3^2}{F_2 + F_3} & 0 \\ -\frac{F_1 F_4}{F_1 + F_4} & 0 & 0 & \frac{F_4^2}{F_1 + F_4} \end{pmatrix}$$

4. BINOMIAL RESPONSE RELATIONS

A further example of the applicability of the general procedure is given by the analysis of a problem presented in Bock (1975, p.508), which is a classical problem in psychophysics of determining the relationship between the discrimination threshold for a stimulus and the level of stimulus intensity at which discriminations are made. Randomly selected subjects were each presented with one of 16 pairs of sucrose solutions in 1-ounce coded containers marked A and B. Subjects tasted the contents of each container in a standard manner and reported the code number of the solution that tasted "sweeter". The observed data is given in Table 4.1.

The relationship considered is a linear or quadratic dependency of the logit frequency on the log of the ratios of sucrose solutions. The relationship is considered separately for each level of B and is given by:

$$(4.1) \quad \text{logit}(F_i) = \log \frac{F_i}{N_i - F_i} = a + b_1 \log \frac{x_A}{x_B} + b_2 \left(\log \frac{x_A}{x_B} \right)^2, \quad i = 1, \dots, 4$$

for the quadratic model, where F_i is the expected number of subjects responding "A sweeter than B" ($A > B$) and x_A and x_B denote the corresponding sucrose concentrations of solutions for A and B respectively.

The values of $\log \frac{x_A}{x_B}$ are approximately $-0,1; -0,033; 0,033; 0,1$ for each level of B. In terms of orthogonal polynomials, the logits of frequencies for each level of B may be modelled as:

(4.2)

$$\text{logit}(\underline{F}) = \begin{pmatrix} 1 & -3 & +1 \\ 1 & -1 & -1 \\ 1 & +1 & -1 \\ 1 & +3 & +1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = X \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

Deviation from this quadratic model is measured by:

$$(4.3) \quad \underline{g}(\underline{F}) = (1, -3, 3, -1) \text{logit}(\underline{F}).$$

Hence if $\underline{F}_0 = (F_{01}, F_{02}, F_{03}, F_{04})'$ and $F_{0i} = F_i(N_i - F_i)/N_i$, $i = 1, \dots, 4$ then

$$\begin{aligned}
 (4.4) \quad \Sigma &= \text{diag}(\underline{F}_0) = D_{F_0} \\
 G &= (1, -3, 3, -1) \cdot D_{F_0}^{-1} \\
 G\Sigma &= (1, -3, 3, -1) \\
 G\Sigma G' &= \frac{1}{F_{01}} + \frac{9}{F_{02}} + \frac{9}{F_{03}} + \frac{1}{F_{04}}.
 \end{aligned}$$

It follows from (2.4) that under quadratic response:

$$(4.5) \quad \underline{F}_c = \underline{F} - \begin{pmatrix} +1 \\ -3 \\ +3 \\ -1 \end{pmatrix} e$$

where

$$(4.6) \quad e = \frac{g(\underline{F})}{1/F_{01} + 9/F_{02} + 9/F_{03} + 1/F_{04}}.$$

The covariance matrix, making use of (2.5) is given by:

$$(4.7) \quad \Sigma_c = D_{F_0} - \frac{1}{1/F_{01} + 9/F_{02} + 9/F_{03} + 1/F_{04}} \begin{pmatrix} +1 \\ -3 \\ +3 \\ -1 \end{pmatrix} (1, -3, 3, -1).$$

In the case of sampling zeros, the corresponding cell frequency may be adjusted by adding a very small quantity. This would lead to approximate MLE's. However the problem of zero counts is not addressed in this paper. (See e.g. Bishop, Fienberg and Holland (1975) and Freeman (1987).)

The linear model is given by

$$(4.8) \quad \text{logit}(\underline{F}) = \begin{pmatrix} 1 & -3 \\ 1 & -1 \\ 1 & +1 \\ 1 & +3 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

and is tested by

$$(4.9) \quad \underline{g}(\underline{F}) = \begin{pmatrix} 1 & -1 & -1 & +1 \\ 1 & -3 & +3 & -1 \end{pmatrix} \text{logit}(\underline{F}) = \underline{0}.$$

Analogous to the quadratic case, the following expressions are obtained for the conditional frequency vector and covariance matrix

$$(4.10) \quad \underline{F}_c = \underline{F} - \begin{pmatrix} +1 \\ -1 \\ -1 \\ +1 \end{pmatrix} e_1 - \begin{pmatrix} +1 \\ -3 \\ +3 \\ -1 \end{pmatrix} e_2$$

where

$$(4.11) \quad \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \frac{1}{g_{11}g_{22} - g_{12}^2} \begin{pmatrix} g_{22} & -g_{12} \\ -g_{12} & g_{11} \end{pmatrix} \underline{g}(\underline{F})$$

$$\begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} =$$

$$\begin{pmatrix} 1/F_{01} + 1/F_{02} + 1/F_{03} + 1/F_{04} & 1/F_{01} + 3/F_{02} - 3/F_{03} - 1/F_{04} \\ 1/F_{01} + 3/F_{02} - 3/F_{03} - 1/F_{04} & 1/F_{01} + 9/F_{02} + 9/F_{03} + 1/F_{04} \end{pmatrix}$$

and

$$(4.12) \quad \Sigma_c = D_{F_0} - \frac{1}{g_{11}g_{22} - g_{12}^2} \begin{pmatrix} +1 & +1 \\ -1 & -3 \\ -1 & +3 \\ +1 & -1 \end{pmatrix} \begin{pmatrix} +g_{22} & -g_{12} \\ -g_{12} & +g_{11} \end{pmatrix} \begin{pmatrix} 1 & -1 & -1 & +1 \\ 1 & -3 & +3 & -1 \end{pmatrix}.$$

The estimates of the frequencies under linear and quadratic response are given in Table 4.1.

TABLE 4.1
JUDGEMENTS OF THE RELATIVE SWEETNESS
OF PAIRED SAMPLES *A* AND *B* OF VARIOUS
CONCENTRATIONS OF SUCROSE SOLUTIONS

<i>Sucrose solutions</i> <i>g/100 ml</i>		<i>Number of</i> <i>responding</i>	<i>subjects</i> <i>A > B</i>		<i>Total</i>
<i>A</i>	<i>B</i>	<i>Observed</i>	<i>Estimated</i> <i>Linear</i>	<i>Quadratic</i>	
1,190	1,5	6	5,0765	5,6054	23
1,389		6	7,6972	7,1838	21
1,620		12	11,3760	10,8162	21
1,890		15	14,8503	15,3946	21
2,397	3,0	12	10,0516	11,0072	24
2,778		8	12,0483	10,9784	22
3,240		17	14,7488	14,0216	22
4,260		13	13,1514	13,9928	17
6,344	8,0	5	3,7127	4,8965	19
7,408		6	7,3944	6,3104	21
8,640		11	12,0731	10,6896	22
10,080		18	16,8198	18,1035	23
15,860	20,0	5	7,8675	5,5419	21
18,520		14	9,8954	12,3743	19
21,600		13	12,6065	14,6257	19
25,200		14	15,6305	13,4581	20

The estimators of the parameters are given by:

$$(4.13) \quad (X'X)^{-1}X' \logit(\underline{f}_c)$$

and the covariance matrix of the estimators by:

$$(4.14) \quad (X'X)^{-1}X'D_{F_0}^{-1}\Sigma_c D_{F_0}^{-1}X(X'X)^{-1}$$

where the design matrix X is as in (4.2) for the quadratic model and as in (4.8) for the linear model.

The estimates for the parameters and their standard errors are given in Table 4.2.

TABLE 4.2
ESTIMATES OF COEFFICIENTS
(AND STANDARD ERRORS)

<i>B</i>	<i>Coeff</i>	<i>Linear</i>	<i>Model</i>	<i>Quadratic</i>	<i>Model</i>	<i>Z - value</i>
1, 5	α	-0,1899	(0,2333)	-0,1790	(0,2346)	0,50
	β	0,3572	(0,1087)	0,3571	(0,1084)	
	γ			0,1179	(0,2346)	
3	α	0,4506	(0,2345)	0,4829	(0,2425)	0,84
	β	0,2594	(0,1083)	0,2839	(0,1166)	
	γ			0,2029	(0,2412)	
8	α	-0,2070	(0,2398)	-0,1629	(0,2424)	1,19
	β	0,4027	(0,1145)	0,3942	(0,1139)	
	γ			0,2878	(0,2421)	
20	α	0,3811	(0,2424)	0,3818	(0,2490)	-2,14*
	β	0,2978	(0,1101)	0,2912	(0,1092)	
	γ			-0,5340	(0,2494)	

The values in the above table correspond to those given in Bock (1975, p.520), although in this case the results for both models are given for each level of B .

A test statistic for deviation from linearity in model (4.2) is then given by

$$(4.15) \quad z = \hat{\gamma} / \sqrt{\text{var} \hat{\gamma}}$$

which is asymptotically $N(0,1)$ if $\gamma = 0$.

As can be seen from Table 4.2, the test for a zero coefficient of the quadratic logistic function is only rejected at level 4 of B (20 g/100 ml).

The estimators of the parameters in the log-linear model are given by:

$$(5.5) \quad \hat{\lambda}_c = (A'A)^{-1} A' \ln \underline{f}_c$$

and the covariance matrix of the estimators by

$$(5.6) \quad (A'A)^{-1} A' D_{F_c}^{-1} \Sigma_c D_{F_c}^{-1} A (A'A)^{-1}.$$

Since the estimators for the conditional frequencies are the same for all three sampling models, the estimators for the λ parameters are also the same. The effect of a different set of marginal totals, in product multinomial sampling, on the estimates of the λ parameters, can be seen from (5.5). If the marginals for the first row variable, with two categories, are assumed fixed at different values then the row frequency vector can be written as

$$(5.7) \quad \underline{f}^N = \begin{pmatrix} \underline{f}_1^0 \cdot k_1 \\ \underline{f}_2^0 \cdot k_2 \end{pmatrix}$$

so that

$$(5.8) \quad \hat{\lambda} = (A'A)^{-1} A' \ln \underline{f}^0 + (A'A)^{-1} \begin{pmatrix} \ln k_1 k_2 \\ \ln k_1 / k_2 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

and therefore a constant is added to the estimates of the mean and the main effect for the row variable, but all other parameter estimates are unaffected.

Although the conditional covariance matrices of the frequency vectors are not equal for the three sampling schemes, in the case of the covariance matrix of the estimators of the parameters, these differences reduce to a difference in the variance of the mean between multinomial and Poisson sampling, and a difference between the variances of the mean and the parameter(s) corresponding to the fixed marginals for product multinomial

and the other two sampling models. The difference in the covariance matrix for the Poisson and multinomial models is due to the term:

$$\begin{aligned} \frac{1}{N}(A'A)^{-1}A'D_F^{-1}\underline{F}\underline{F}'D_F^{-1}A(A'A)^{-1} &= \frac{1}{N}(A'A)^{-1}A'\underline{1}\underline{1}'A(A'A)^{-1} \\ &= (A'A)^{-1} \begin{pmatrix} m^2/N & \underline{0}' \\ \underline{0} & 0 \end{pmatrix} (A'A)^{-1} \\ &= \begin{pmatrix} \frac{1}{N} & \underline{0}' \\ \underline{0} & 0 \end{pmatrix} \end{aligned}$$

where m is the dimension of the frequency vector. The only parameter affected, therefore, is $\text{var}(\hat{\mu})$. In the same way it can be shown that if, for example, the marginals of a variable are fixed, then only the variances of $\hat{\mu}$ and of the main effects of that variable will differ in the product multinomial, multinomial and the Poisson models. What is important is that the variances of the parameters of interest, such as the interaction terms, are equal in the three sampling models.

6. THE 2^3 TABLE

As an illustration of the theory presented in Section 5 consider a simple $2 \times 2 \times 2$ contingency table. Certain equations are derived in the case of such a table for three different models. The model (5.1) reduces to

$$\begin{aligned} (6.1) \quad \ln \underline{F} = A\underline{\lambda} &= (\underline{1}, \underline{a}_1, \underline{a}_2, \dots, \underline{a}_7)\underline{\lambda} \\ &= \begin{pmatrix} 1 & +1 & +1 & +1 & +1 & +1 & +1 & +1 \\ 1 & +1 & +1 & -1 & +1 & -1 & -1 & -1 \\ 1 & +1 & -1 & +1 & -1 & +1 & -1 & -1 \\ 1 & +1 & -1 & -1 & -1 & -1 & +1 & +1 \\ 1 & -1 & +1 & +1 & -1 & -1 & +1 & -1 \\ 1 & -1 & +1 & -1 & -1 & +1 & -1 & +1 \\ 1 & -1 & -1 & +1 & +1 & -1 & -1 & +1 \\ 1 & -1 & -1 & -1 & +1 & +1 & +1 & -1 \end{pmatrix} \begin{pmatrix} \mu \\ \lambda^A \\ \lambda^B \\ \lambda^C \\ \lambda^{AB} \\ \lambda^{AC} \\ \lambda^{BC} \\ \lambda^{ABC} \end{pmatrix} \end{aligned}$$

In the saturated model the estimator and estimated variance of $\hat{\lambda}^{ABC}$ is given by

$$(6.2) \quad \hat{\lambda}^{ABC} = \underline{a}_7' \ln \underline{f} / \underline{a}_7' \underline{a}_7 = \frac{1}{8} \underline{a}_7' \ln \underline{f}$$

$$(6.3) \quad \text{var } \hat{\lambda}^{ABC} = \underline{a}'_7 D_F^{-1} \underline{a}_7 / (\underline{a}'_7 \underline{a}_7)^2 = \frac{1}{64} \underline{a}'_7 D_F^{-1} \underline{a}_7.$$

For the model in which $\lambda^{ABC} = 0$, the conditional frequency vector is

$$(6.4) \quad \underline{F}_c = \underline{F} - e \cdot \underline{a}_7$$

where

$$e = \underline{a}'_7 \ln \underline{F} / \underline{a}'_7 D_F^{-1} \underline{a}_7.$$

The conditional covariance matrix of $\hat{\underline{F}}_c$ is given by

$$(6.5) \quad \Sigma_c = \Sigma - \underline{a}_7 \underline{a}'_7 / \underline{a}'_7 D_F^{-1} \underline{a}_7.$$

It follows, for example, that

$$(6.6) \quad \lambda_c^{BC} = \underline{a}'_6 \ln \underline{F}_c / \underline{a}'_6 \underline{a}_6 = \frac{1}{8} \underline{a}'_6 \ln \underline{F}_c$$

and

$$(6.7) \quad \begin{aligned} \text{var}(\hat{\lambda}_c^{BC}) &= \underline{a}'_6 D_{F_c}^{-1} \underline{a}_6 / (\underline{a}'_6 \underline{a}_6)^2 \\ &\quad - (\underline{a}'_6 D_{F_c}^{-1} \underline{a}_7)^2 / \underline{a}'_7 D_{F_c}^{-1} \underline{a}_7 \cdot (\underline{a}'_6 \underline{a}_6)^2 \\ &= \frac{1}{64} \left\{ \underline{a}'_6 D_{F_c}^{-1} \underline{a}_6 - \frac{(\underline{a}'_6 D_{F_c}^{-1} \underline{a}_7)^2}{\underline{a}'_7 D_{F_c}^{-1} \underline{a}_7} \right\}. \end{aligned}$$

If B and C are independent, it implies that $\lambda^{ABC} = \lambda^{BC} = 0$. The conditional frequency vector now becomes

$$\begin{aligned} \underline{F}_c &= \underline{F} - (\underline{a}_6, \underline{a}_7) \begin{pmatrix} \underline{a}'_6 D_{F_c}^{-1} \underline{a}_6 & \underline{a}'_6 D_{F_c}^{-1} \underline{a}_7 \\ \underline{a}'_7 D_{F_c}^{-1} \underline{a}_6 & \underline{a}'_7 D_{F_c}^{-1} \underline{a}_7 \end{pmatrix}^{-1} \begin{pmatrix} \underline{a}'_6 \ln \underline{F}_c \\ \underline{a}'_7 \ln \underline{F}_c \end{pmatrix} \\ &= \underline{F} - e_1 \underline{a}_6 - e_2 \underline{a}_7 \end{aligned}$$

where

$$\begin{aligned} e_1 &= \{ \underline{a}'_7 D_{F_c}^{-1} \underline{a}_7 \cdot \underline{a}'_6 \ln \underline{F}_c - \underline{a}'_6 D_{F_c}^{-1} \underline{a}_7 \cdot \underline{a}'_7 \ln \underline{F}_c \} / \det \\ e_2 &= \{ \underline{a}'_6 D_{F_c}^{-1} \underline{a}_6 \cdot \underline{a}'_7 \ln \underline{F}_c - \underline{a}'_6 D_{F_c}^{-1} \underline{a}_7 \cdot \underline{a}'_6 \ln \underline{F}_c \} / \det \\ \det &= \underline{a}'_6 D_{F_c}^{-1} \underline{a}_6 \cdot \underline{a}'_7 D_{F_c}^{-1} \underline{a}_7 - (\underline{a}'_6 D_{F_c}^{-1} \underline{a}_7)^2. \end{aligned}$$

Consider the example of Bartlett's data given in Bishop, Fienberg and Holland (1975, p. 87). The estimates of the frequencies obtained from the general procedure proposed, are given in Table 6.1 for the two unsaturated models considered. The values for each iteration of the procedure are contained in the table and the constant value(s) to be subtracted from each frequency have also been given.

TABLE 6.1
FREQUENCY ESTIMATES FOR BARTLETT'S DATA
MODEL $\lambda^{ABC} = 0$

Cell CBA	Observed value	Iterations	
		f_1	f_2
111	156	161,0007	161,0961
211	84	78,9993	78,9039
121	84	78,9993	78,9039
221	156	161,0007	161,0961
112	107	101,9993	101,9039
212	133	138,0007	138,0961
122	31	36,0007	36,0961
222	209	203,9993	203,9039
e	-5,0007	0,0954	0,0000

$$\hat{\lambda}^{ABC} = -0,057, \text{ var}(\hat{\lambda}^{ABC}) = 0,0014, \text{ standardized } \hat{\lambda}^{ABC} = -1,52$$

$$\text{MODEL } \lambda^{ABC} = \lambda^{BC} = 0$$

Cell CBA	Observed value	Iterations		
		f_1	f_2	f_3
111	156	122,2005	120,0005	120
211	84	117,7995	119,9995	120
121	84	117,7995	119,9995	120
221	156	122,2005	120,0005	120
112	107	75,6346	69,0312	69
212	133	164,3654	170,9688	171
122	31	62,3654	68,9688	69
222	209	177,6346	171,0312	171
e_1	32,5825	4,4017	0,0158	0
e_2	1,2171	-2,2017	-0,0153	0

$$\hat{\lambda}^{BC} = 0,357, \text{ var}(\hat{\lambda}^{BC}) = 0,0013, \text{ standardized } \hat{\lambda}^{BC} = 9,90$$

The final estimates given in the above table are the same as those given by the iterative proportional fitting procedure.

7. A THREE-WAY EXAMPLE

As an illustration we consider the example given in Haberman (1978, p.183). The results of fitting a model in which the parameter $\lambda^{SER} = 0$ to the $2 \times 3 \times 2$ table using (5.3) (two iterations) and using the Newton-Raphson algorithm (three iterations) are given in Table 7.1.

The corresponding computations making use of the iterative proportional fitting (IPF) procedure are given in detail in Haberman (1978, p.195). In this example the estimation procedure given by expression (5.3) is more easily computable and converges faster than the Newton-Raphson or IPF procedures. Comparing the merits of different computing procedures is of course a difficult task, since other criteria like programming complexity and storage space should also be taken into account.

TABLE 7.1

COMPUTATION OF FREQUENCY ESTIMATES
FOR THE MODEL OF NO THREE-WAY INTERACTION

Variables SER	Observed value	Newton - Raphson			(5.3)	
		f_1	f_2	f_3	f_1	f_2
111	72	77,186	77,048	77,051	76,984	77,051
112	47	42,498	41,952	41,949	42,016	41,949
121	110	112,356	112,640	112,644	112,535	112,644
122	196	192,791	193,352	193,356	193,465	193,356
131	44	37,352	36,320	36,305	36,482	36,305
132	179	186,019	186,689	186,695	186,518	186,695
211	86	81,146	80,947	80,949	81,016	80,949
212	38	43,557	43,052	43,051	42,984	43,051
221	173	169,935	170,352	170,356	170,465	170,356
222	283	284,269	285,637	285,644	285,535	285,644
231	28	36,872	35,711	35,695	35,519	35,695
232	187	179,019	179,301	179,305	179,482	179,305
$\ f_{i+1} - f_i\ $		2,4480	0,0255	0,0000	0,4363	0,0000

8. THE ANALYSIS OF FREQUENCY STRUCTURES

The estimation procedure may be applied to the analysis of frequency structures in a straight-forward way. These include e.g. frequency tables with structural zeros or block-triangular forms.

Consider as a simple example the example given in Bishop, Fienberg and Holland (1975, p. 288) where Merino ewes are cross-classified by the number of lambs born to them in two consecutive years, 1952 and 1953. (See Table 8.1). The number of lambs born to an ewe in 1952 is related to the number in 1953 and the problem is that of testing quasi-independence, given the diagonals fixed. Under Poisson sampling the only degree of freedom for independence which is orthogonal to the structure of fixed diagonals, is represented by:

$$(8.1) \quad g(\underline{F}) = \underline{a}' \ln \underline{F}$$

where $\underline{a}' = (0, 1, -1, -1, 0, 1, 1, -1, 0)$ and

$$(8.2) \quad \Sigma = \text{diag}(0, F_2, F_3, F_4, 0, F_6, F_7, F_8, 0).$$

Hence

$$(8.3) \quad \underline{F}_c = \underline{F} - e\underline{a}$$

with $e = \underline{a}' \ln F / (F_2^{-1} + F_3^{-1} + F_4^{-1} + F_6^{-1} + F_7^{-1} + F_8^{-1})$. The results are given in Table 8.1.

TABLE 8.1
CROSS-CLASSIFICATION OF EWES ACCORDING TO NUMBER OF
LAMBS BORN IN CONSECUTIVE YEARS

Number 1953	Number 1952	Observed value	Iterations	
			f_1	f_2
0	0	58	58	58
	1	52	51,133	50,987
	2	1	1,867	2,013
1	0	26	26,867	27,013
	1	58	58	58
	2	3	2,133	1,987
2	0	8	7,133	6,987
	1	12	12,867	13,013
	2	9	9	9
e		0,8668	0,14562	0,0002

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